The Adaptive Reflex Engine
V1.7 — Multi-Reflex Chains at Million Scale

A detailed report on the nineteenth experiment of the arc: what was built, what four frontier models did with it, what four million evaluations proved, where it broke, and where the programme stands.

September 4, 2026 · repo **reflex-engine**, tag **v1.7-final** (819115d), frozen snapshot **reflex-engine-v1.7-frozen** · spec frozen before any sealed run (v1.7-plan, b87ff8f); sealed datasets hashed and evaluated once.

Headline numbers

What	Result
Total evaluations	4,000,000 (1M per model arm) + 700k design/control runs; zero uncontained engine errors
Deterministic chain serving (sealed 300k)	112,170 answers, 0 wrong, 0 narrative traps certified
3-6-step questions solved deterministically	30.4% of the 3+-step sealed stratum, at zero wrong
Sealed plans certified (1,200/model)	opus 1,150 | fable 1,138 | sonnet 1,118 | haiku 1,077
Certified-wrong plans	haiku 13 | sonnet 0 | fable 0 | opus 0
Structure-finding time (median)	4.60-6.45 s; +10-19% from 3-step to 6-step chains
Wrong answers in the four 1M arms	haiku 3,069 | sonnet 2,768 | fable 1,762 | opus 1,243 (of ~0.46-0.52M serves each; every one
Pre-registered outcome	E on the frozen ladder (safety-first letter); every capability criterion passed

1. The concept: what this programme is trying to build

Large language models answer any question, including wrongly, with the same fluent confidence. The Adaptive Reflex Engine is a nineteen-experiment attempt to build the opposite kind of system: one that answers **only when it can prove the answer** and says UNKNOWN otherwise. The design principle of the whole arc is **UNKNOWN > WRONG**: an abstention costs a fallback to a slower path; a confidently wrong answer poisons everything downstream.

The unit of capability is a **reflex**: a small, typed, deterministic procedure that recognizes one shape of question ("kinetic energy of a mass at a speed", "lcm of two integers", "first derivative at a point") and computes the answer with ordinary code, not with a neural network. Three properties make reflexes worth growing into a library:

- **Certifiability.** A reflex's answer is checked by an *independent witness* - a second implementation on a different math library (mpmath vs sympy) that must agree before anything is served. Two independent computations agreeing by accident is how the arc gets its zero-wrong-fire records.
- **Learnability.** Models don't answer questions in this architecture - they *propose procedures and structure*, which then face a gauntlet (probation on held-out traffic, zero tolerance for a wrong serve) before admission. The model is a miner; the engine is the assayer.
- **Economy.** A reflex serve costs milliseconds of CPU; an LLM call costs seconds and joules. If a growing library absorbs a rising share of traffic, the fleet-level economics change - that was the original commercial premise (see the V1.4 caveat in section 7).

By V1.6 the engine certified **single** scientific facts and calculations at scale with literally zero wrong answers across 118k+ sealed serves. V1.7 asked the composition question: real questions are chains - *"integrate this, use the result as a speed, compute the kinetic energy, convert to kJ"* - so can certified reflexes be **chained**, can models **plan** such chains, and how long does the planning take?

How every claim in this report is protected

- **Pre-registration.** Hypotheses, metrics, thresholds and an outcome ladder (A best - F worst) are frozen in the spec before any sealed data is touched. The outcome letter is decided by the frozen text, not by post-hoc judgement.
- **Sealed evaluation.** Datasets are hashed at freeze; sealed splits are run once per pre-registered arm; dev and sealed never mix.
- **Bug audit.** Every defect gets a numbered entry (B1-B32 across the arc), superseded runs are preserved in-repo, and negative results are reported with the same prominence as positive ones.
- **Harness-observed timing.** Model self-reports are never trusted for time or energy (bug B7); V1.7 timing comes from file-modification timestamps observed by the harness.

2. Eighteen experiments of context: how we got here

The arc alternates between capability pushes and honest failures; roughly every second version kills one of its own premises. The chart below shows the pre-registered outcome of each version and its defining result.

The arc at a glance: pre-registered outcome letters (A best - F worst) and each version's defining result

A	**V0.5**	574 vs 1687 J/correct
E	**V0.7**	recognition wall
C	**V0.8**	typed boundary = the mechanism
F	**V0.9**	depth wall at 5-7 steps
F	**V0.10**	196/196 at 0 LLM calls; gate excluded all
A	**V0.11**	0.000 wrong-exec everywhere
C	**V1.0**	acquisition channel proven (+7.2pp)
A	**V1.1**	generator universe closed; 95.5% @ 0 wrong
B	**V1.2**	certified 0.000 over 10,160 traps / 65 seeds
D	**V1.3**	real language read: 209 certified / 3 wrong
D	**V1.4**	energy premise inverted (19.9x GPU)
E	**V1.5**	composition is the bottleneck
B	**V1.6**	foreign coverage 35.4 -> 55.5% @ 0 wrong
E	**V1.7**	chains compose: 30.4% @ 0 wrong; reuse unsafe

Fig. 1 - the arc V0.5-V1.7. Outcome letters are decided by each version's pre-registered ladder; E/F are safety or premise failures, deliberately prioritized over capability wins.

Milestones that shaped V1.7 directly:

- **V0.9-V0.10 - the depth wall.** Early planners assembled 5-7-step paths at 0/25 success while shallow paths were perfect 125/125. Composition, not recognition, was flagged as the hard problem seven experiments early.
- **V1.1 - typed reflexes at commercial scale.** 100k-query simulation: 95.5% certified at 0 wrong. Also the first *scale inversion* scare - the largest miner minted more wrong candidates - later shown to be architectural, not a model law.
- **V1.2 - formal certification.** 10,160 adversarial traps, 65 seeds, 0.000 wrong-fire, machine-checked certificate. Also first contact with real language: 519/520 abstentions - safe and useless.
- **V1.3 - the sentence that named V1.7's villain.** Plan reflexes were falsified with 7 served / 7 wrong: *checks verify a plan matches the TEXT, never that it answers the QUESTION.* V1.7 rebuilt planning specifically to beat this, and its one surviving failure mode (section 6) is this sentence's echo.
- **V1.4 - the energy inversion.** The measured reflex pipeline of that era cost 19.9x more GPU energy per query than a pure LLM while answering 0/60 vs 51/60. The commercial premise was inverted, not merely unproven; everything since is justified on safety and correctness, with economics an open question.
- **V1.5 - composition named the bottleneck.** Knowing a story contains 47, 41 and 20 is not knowing the answer is 47-41+20. 40% wrong-fire on sealed compositions; guards encode which compositions are wrong, never which is right.
- **V1.6 - the re-scope that worked.** Scoped to scientific/arithmetic/logical questions only, four-model matrix (haiku/sonnet/opus/fable), watched minting, 47-leaf taxonomy: 12 sealed runs, 118k+ serves, ZERO wrong-fire; foreign-phrased coverage grown 35.4% -> 55.5% by a 271-reflex minted library; V1.1's inversion falsified (quality is monotone in scale).

3. What V1.7 built

The user brief: ~one million queries per model; every question requires one to six reflexes chained through each other; measure cohesiveness, combined calculation, and how long each model takes to find and implement a multi-reflex structure; let every model mint freely, including a new *connector* class that joins two existing reflexes.

3.1 The data: Q-1M

1,000,000 generated chain questions (seed 1700, committed in-repo with raw-byte hashes): 60.6% non-monotonous phrasing, 71.7% span multiple domains (calculus into finance into physics), 6,000 foreign-phrased chains, 3,624 narrative distractors that must NEVER be answered. Depth mix k=2:20%, k=3:30%, k=4:25%, k=5:15%, k=6:10%. Splits: ACQ 200k (minting offers), DEV 100k (tuning allowed), SEAL 300k (run once), REPLAY 400k (reuse economics).

3.2 The machinery

- **Plan schema** - a typed DAG: nodes like `evaluate`, `formula`, `derivative_at`, `integrate_def`, `solve_pick`, `convert`, `ncr`, `probability`; later nodes reference earlier outputs as `@n1`. A plan is data, not prose - executable and auditable.
- **Composer** - the deterministic chain parser: reads a chain question, emits a plan, `role-aware` (it knows 7% is a rate and 7 years is a duration because it parses the roles, not positions).
- **Gate G-P1..4** - static plan checks: schema/DAG sanity; type and unit flow (you cannot feed joules into `is_prime`); ask-target (the final node must answer the asked quantity in the asked unit); stated-number honesty (every number the question states must be used; no number the question doesn't state may appear, up to a small whitelist of structural constants like 0.5, 100, 9.8).
- **Dual execution** - the sympy-side executor's every step and final value must be reproduced by an independent mpmath-only witness with depth-budgeted tolerance. Control batteries (one sound plan + eight wrong-plan classes; witness positive controls) passed 100% before any counting run.
- **Three minted classes** - parse reflexes (V1.6 class), chain reflexes (pattern + plan template), and the user-proposed connector class (unit adapters joining two reflexes). All face chained probation: at least 3 end-witnessed correct serves, at least 2 distinct operand tuples, and one wrong serve kills the candidate outright.

The arms. `det` = composer + frozen V1.6 library, no model content. `M-` = det + that model's gauntlet-admitted reflexes. `R-` = det + minted reflexes + that model's dev-phase plan templates, run over the full million. Planning campaigns S-PLAN: each model structures 600 dev + 1,200 sealed escalated questions, one at a time, timed.

4. Result 1 - reflexes compose: the deterministic engine on chains

The sealed 300k, run once: **112,170 answers served, zero wrong, zero of 1,095 narrative distractors certified**. In-domain coverage 37.5% overall; on the stratum that defines the version - questions needing three or more chained steps - coverage is **30.4%** (72,793 of 239,299), against a pre-registered 25% cohesion bar.

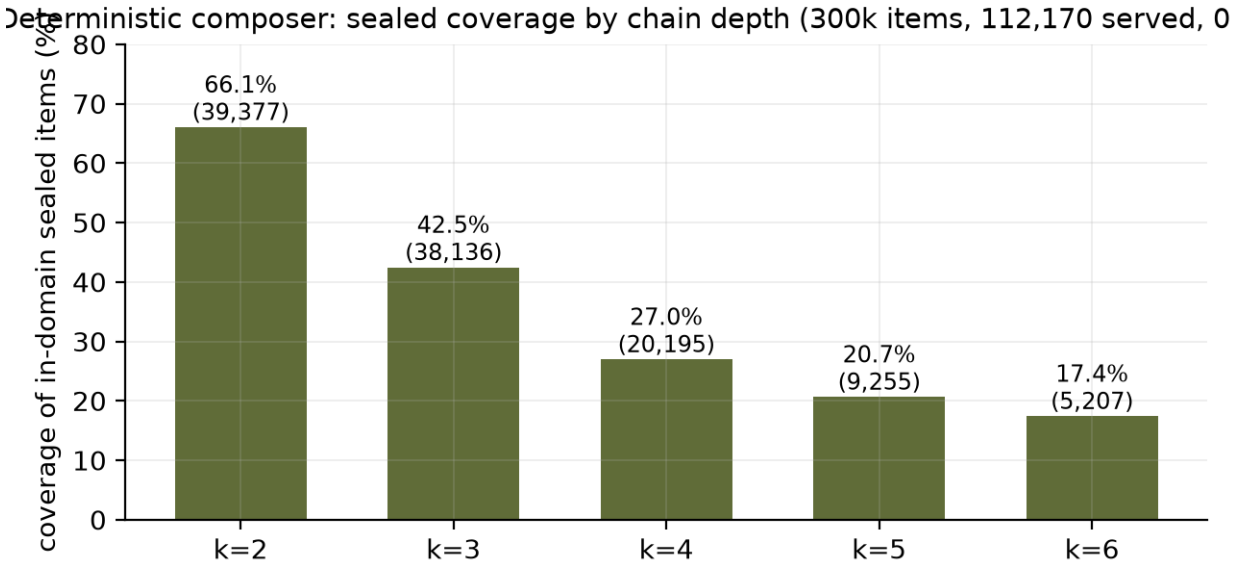


Fig. 2 - deterministic sealed coverage by depth. Decay is graceful (each extra step multiplies roughly 0.6-0.75x), not a wall: V0.9's 0/25 cliff beyond 5 steps is gone - k=6 chains serve at 17.4%, every one witnessed end-to-end.

Two structural rules keep this number honest. A **partial answer to a chain is a wrong answer** (bug B30, section 6) - chain-shaped text is banned from the single-step engine entirely. And a minted chain pattern must **consume the whole question** (bug B31) - no reflex may match a prefix and serve the two-step answer to a four-step question.

5. Result 2 - models plan chains: 4,800 sealed structures, watched and timed

Each model received 1,200 sealed questions the deterministic engine had escalated - the residual it could NOT do - in 25-question batches, one serial lane per model. The model sees the question, writes the plan as a JSON DAG, one file per item; the harness measures the write-time gaps. Every plan then faces the full gate and the dual witness. The model only ever proposes structure; the engine computes every number.

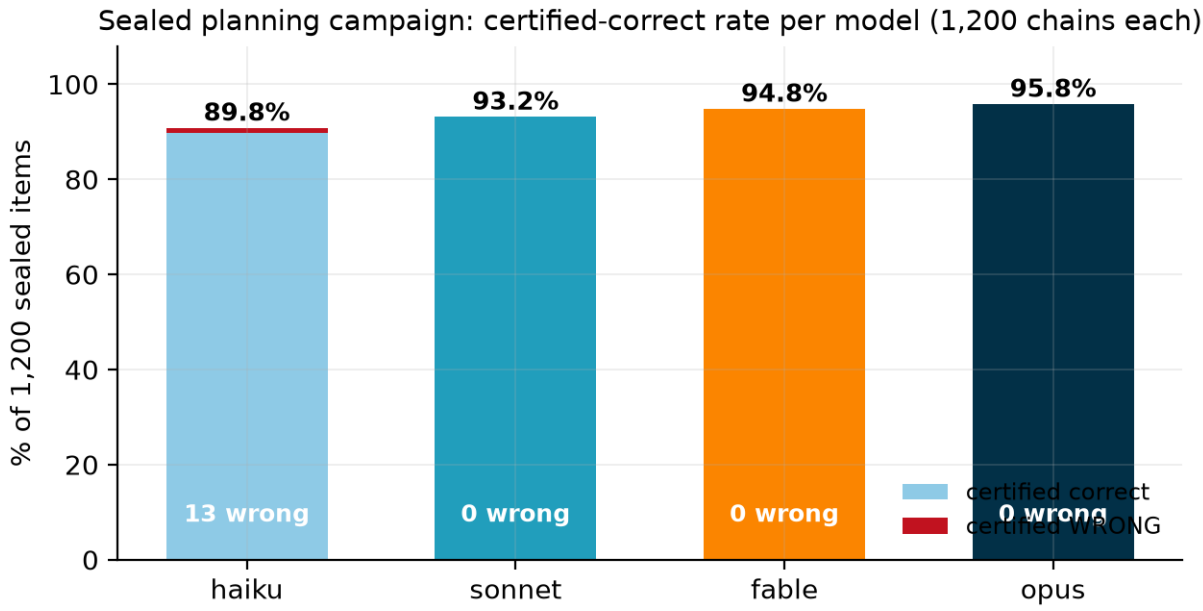


Fig. 3 - sealed certified-correct rate. Certified = passed G-P1..4, executed, and independently witnessed. haiku's red sliver is the 13 certified-wrong plans discussed in section 6.

model	certified	rate	gate-killed	exec-refused	certified-WRONG	dev wrongs (of 600)
haiku	1,077	89.8%	81	29	13	0
sonnet	1,118	93.2%	66	16	0	0
fable	1,138	94.8%	62	0	0	0
opus	1,150	95.8%	50	0	0	0

Three of four frontier models structured ~1,150 multi-step scientific questions each - questions the deterministic parser had already failed - with **not one wrong certified answer**. The dev column carries a methodological lesson: 600-item dev samples showed zero wrongs for ALL four models; only the 1,200-item sealed set surfaced haiku's 13. Rare failure modes need sealed volume, not dev iterations.

5.1 The timing question

The brief asked directly: how long does an LLM take to figure out and implement a multi-reflex structure?

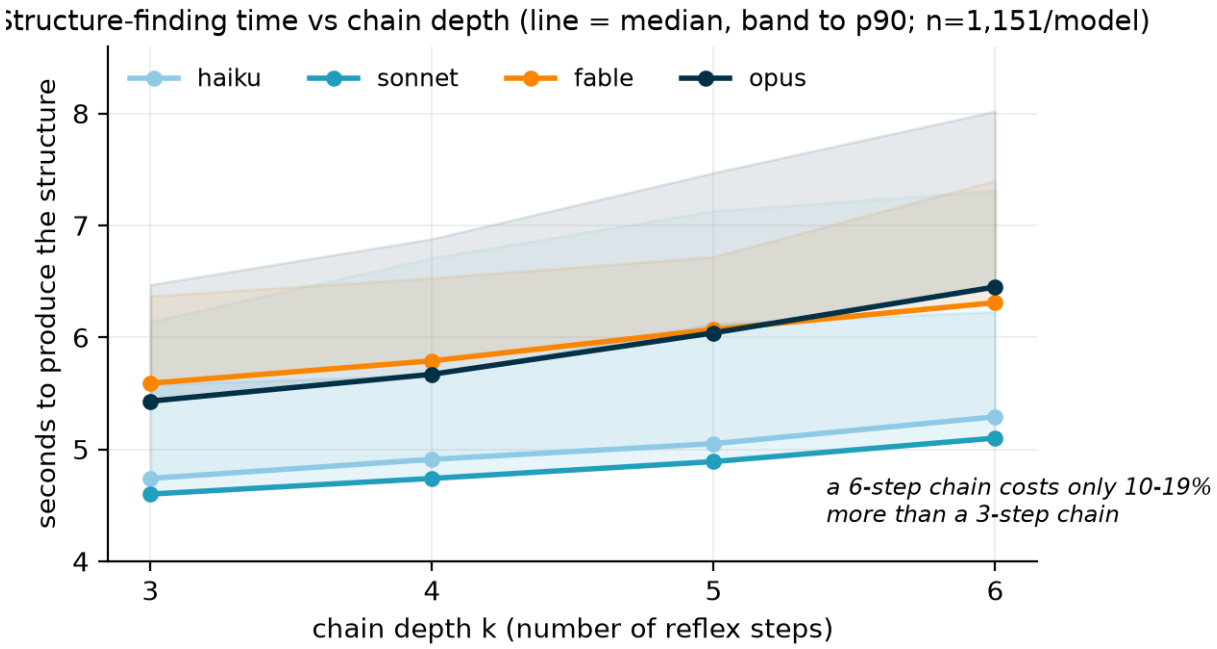


Fig. 4 - median structure-finding time per chain, with p90 band. Harness-observed file mtimes; first item of each 25-item batch excluded as agent startup; 1,151 of 1,200 items measured per model (95.9%).

• **Depth is almost free.** A 6-step structure costs 4.7-19% more than a 3-step one (sonnet +0.50s, haiku +0.55s, fable +0.72s, opus +1.02s). Models read the whole chain in one pass; they do not re-derive step by step. This falsifies the intuition that structure-finding time compounds with depth.

• **Speed anti-correlates with safety.** The two fastest planners (sonnet 4.60s, haiku 4.74s at k=3) bracket the only unsafe one; the slowest, opus (6.45s at k=6), is the most accurate at zero wrong. Fable sits with opus: slower, zero-wrong, with the tightest p90 tail of the four (6.72s at k=5 vs haiku's 7.13s).

• **Cost intuition:** at ~5s and zero marginal engine cost per certified structure, one plan certifies a *class* - see reuse economics (eta) in section 5.2 - but reuse is exactly where V1.7 broke; the two must be read together.

5.2 Lift and reuse economics

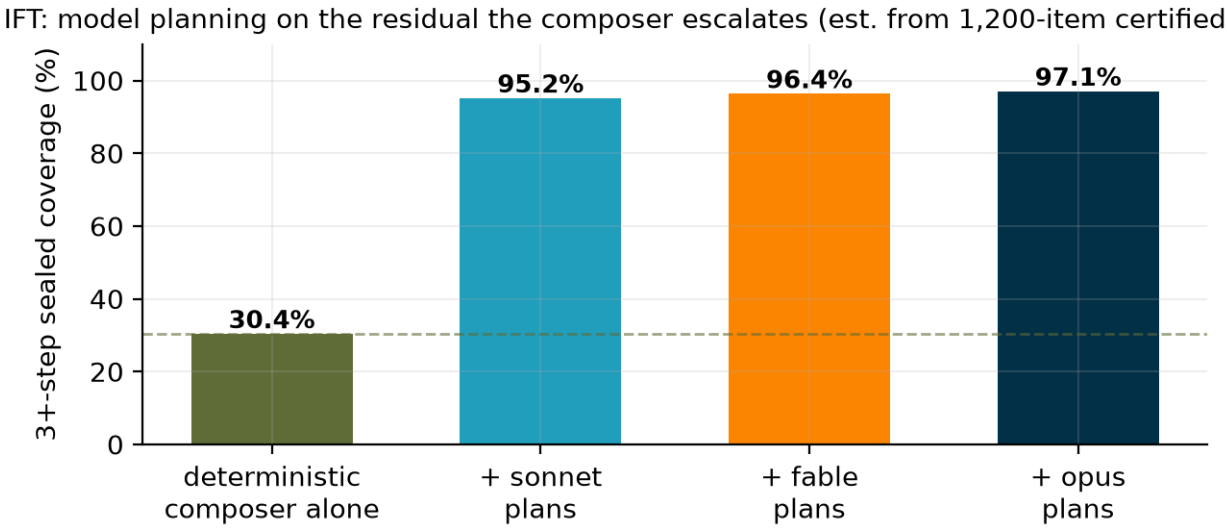


Fig. 5 - what model planning adds on the 3+-step sealed stratum. Bars for models are estimates: certified rate on the 1,200-item uniform sample of the escalation pool, extrapolated over the 166,506-item residual; all at zero certified-wrong.

The deterministic composer alone covers 30.4% of 3+-step sealed questions; adding a large model's certified planning takes the stratum to an estimated 95-97% - roughly a **two-thirds jump (+64.8 to +66.7 points)** at zero wrong for sonnet, fable, and opus. Reuse mass: on the 400k replay split, each admitted library entry served on average haiku 12.5 / sonnet 48.2 / fable 65.7 / opus 74.9 times - one ~5-second planning act amortizes across dozens of serves. That economics claim, however, is exactly what section 6 wounds.

6. Where it broke - exact counts, no rounding away

V1.7's negative result is as load-bearing as its positive one. Everything below is measured, committed, and traced row by row.

6.1 The million arms: template replay served wrong answers

The R-arms carried each model's dev plans forward as reusable *templates*: the question's numbers became regex capture groups, the plan's structure was kept. Over the four million items, this replay layer served wrong answers at every scale:

arm	ACQ 200k	DEV 100k	SEAL 300k	REPLAY 400k	total wrong	sealed wrong-fire	error/template serve
R-haiku	605	289	937	1,238	3,069	0.80%	18.7%
R-sonnet	555	261	840	1,112	2,768	0.63%	4.0%
R-fable	345	167	542	708	1,762	0.38%	1.9%
R-opus	256	108	357	522	1,243	0.24%	1.06%

Per-model million arms (R-<model>, 1M items each): every metric improves monotonically with scale

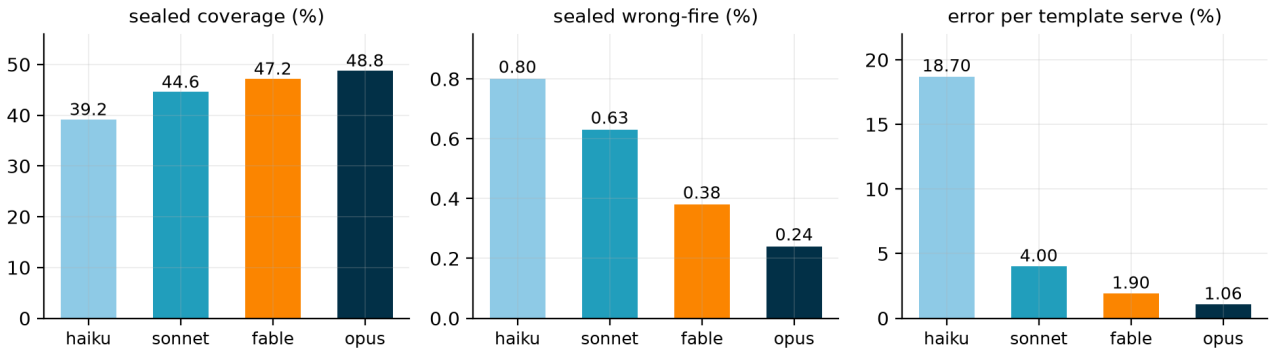


Fig. 6 - the scale gradient across the four 1M arms. Coverage rises and both error measures fall monotonically from haiku to opus; V1.1's feared scale inversion stays falsified one architectural layer up.

Crucially, **the deterministic composer inside those same runs was at zero wrong** - the 112,170 sealed composer serves recur identically in every arm with 0 errors. Every single wrong answer in 4,000,000 evaluations traces to a replayed template (`mint:chain:_plan_*` in the row-level ledgers).

6.2 B32 - the anatomy of the failure

Bug B32 is the version's principal discovery. Template replay binds numbers **by position**: whatever number sits where the donor question's rate sat becomes the rate. Three mechanisms, one family:

- **Donor-constant swap.** Template literals that fall inside the anti-value-smuggling whitelist (rates 2-7%, small multipliers, year counts) travel with the template. A template minted from *'interest at 2% over 6 years, multiply by 3'* matches a sibling question with rate 5% over 8 years - and computes the donor's arithmetic. The honesty gate cannot object: the whitelist exempts exactly those numbers. Example, traced: q17_0346197 served 2.16 where gold was 6.48 - the donor's multiplier x3 dropped.
- **Role misbinding.** The regex groups capture numbers that ALL appear in the question - in the wrong roles: rate and years swapped, a multiplier bound as a bound of integration (one traced case integrated over [1,1] and served force = 0.0 where gold was 6,191.0). No literal-presence check can see this; every literal is legitimately present.
- **Dropped dual-role usage** - haiku's 13 certified-wrong *plans* (not templates). When one stated number serves two roles ('multiply by 7 ... at 7% per year'), a plan that uses 7 once satisfies magnitude-level honesty while silently dropping a step. 12 of the 13 sit on derivative- or sum-headed chains into interest with exactly

this shape.

A structural fix was implemented and probed on dev (never on the run-once sealed data): `--strictreplay` refuses any template serve whose bound numbers are not all stated in the question. On a 5,000-item probe it blocked the donor-constant subset and cut wrong-fire from 0.73% to 0.46% - and **9 wrongs survived**, all role misbindings. The conclusion is not 'filter harder'; it is architectural:

> **Positional reuse is structurally unsafe. The composer is role-aware and had zero wrongs in 412,000+ serves; the templates are role-blind and no presence filter can teach them roles. Safe plan reuse must bind by role - that is V1.8's problem statement.** (V1.3, four experiments ago: 'checks verify a plan matches the TEXT, never that it answers the QUESTION.' The class was beaten for fresh plans by the gate+witness stack; it re-entered through reuse.)

6.3 The other bugs of record

- **B30** - on first contact with merged chain traffic, V1.6's single-step parser answered the FIRST sub-question of 5 chains (wrong-fire 0.008% and 0.003% - the arc's first nonzero since V1.6's dev loop). Fix: chain-marked text is out of the single-step engine's domain *by definition*. Reruns: zero.
- **B31** - one minted sonnet reflex (`chain_lcm_interest`) was not end-anchored and served 2-step answers to 67 4-step dev questions. Probation had never sampled a colliding longer chain; dev's 100k did. Fix: whole-question consumption enforced in the gauntlet AND the serving path; the reflex died by generalization. Reruns: zero.
- **Connectors - an honest null.** All four models proposed connector reflexes (8-12 each); every one was inert in probation because the composer already performs final-unit conversion. Free-hand invention found no gap where one was hypothesized - reported as a result, not omitted.

7. What we have achieved - and what we have not

7.1 Achievements, arc-level

- **A certified computational substrate.** From V1.2's formal certificate (10,160 traps, 65 seeds, 0.000 wrong-fire) through V1.6's 118k+ sealed serves at zero wrong to V1.7's 112,170 sealed *chain* serves at zero wrong: the deterministic layer has not produced a single wrong certified answer since V1.6 froze - across roughly 550,000 sealed serves.
- **Composition solved deterministically to 30.4%** of 3+-step questions - the exact capability V0.9 (depth wall) and V1.5 (composition bottleneck) failed at. Chains of six certified steps now serve with end-to-end independent witnessing.
- **A working division of labour with frontier models.** Models propose procedures (V1.6: 271 admitted reflexes, +20.1pp foreign coverage) and structure (V1.7: 93-96% of the deterministic residual certified at zero wrong by three of four models). The engine computes; the models never touch a number that gets served.
- **Quantified planning behaviour.** First measured answer to 'how long does structure-finding take': ~5 seconds, nearly flat in depth, safety anti-correlated with speed.
- **A reproducible method.** Nineteen pre-registered experiments, 32 numbered bugs, every superseded run preserved, every wrong answer committed as a row. The negative results (V1.4 energy inversion, V1.5 composition failure, B32) are as citable as the positives.

7.2 Limitations, stated plainly

- **Reuse is unsafe.** The template replay layer produced 8,842 wrong answers across 4M evaluations (0.24-0.80% of serves per arm). Until reuse binds by role, every 'plans amortize' economic claim is conditional.

• **haiku plans unsafely at sealed scale.** 13 certified-wrong plans out of 1,090 served (1.19%) - all dual-role number drops. The gate's magnitude-level honesty is provably insufficient for the smallest model; role-aware accounting is needed at the gate too.

• **Coverage ceilings.** Deterministic: 17.4% at k=6; overall 37.5% sealed. The engine still escalates most deep traffic; model planning covers the gap only at ~5s latency per novel structure.

• **Domain scope.** Everything here is scientific/arithmetic/logical calculation - the V1.6 re-scope. Narrative and open-domain question answering remain out of scope; V1.2-V1.5 showed how brutal that frontier is (519/520 abstentions; then 40% wrong-fire when forced).

• **Economics unproven.** V1.4 measured the then-pipeline at 19.9x MORE GPU energy than a pure LLM. V1.7 made no energy claims (per bug-discipline B7) - the compute economics of the current, much leaner engine are an open measurement, not an assumption.

• **The outcome letter is E.** The frozen ladder puts any safety-bar breach above every capability win, and two of four million-arms breached (0.80%, 0.63% vs the 0.5% ceiling), as did haiku's plan arm. We report the letter with the numbers because the discipline of losing the letter honestly is what makes the zero-wrong claims elsewhere believable.

8. Where we stand

The system today: a deterministic engine that answers single-step scientific questions at ~56% (foreign-phrased) to 95%+ (native register) and multi-step chains at 37.5%, with a zero-wrong record across every sealed serve since V1.6; a model-planning channel that certifies most of the remaining chain questions at zero wrong for sonnet-class models and above, at ~5 seconds per novel structure; and a quarantined reuse layer whose failure mode is now precisely characterized down to the individual misbound number.

The single most consequential sentence of V1.7: **in four million evaluations, no wrong answer ever came from a role-aware component - and every wrong answer came from a role-blind one.** That converts V1.8 from an open search into an engineering task: give plan reuse the same role-aware binding the composer already has (match templates by structure signature AND role slots, not positional regex; extend stated-number honesty from magnitudes to role-tagged usages, which also closes haiku's dual-role class). The bet is falsifiable and will be pre-registered like the eighteen before it.

Artifacts: tag v1.7-final (819115d); frozen snapshot reflex-engine-v1.7-frozen; Q-1M with raw-byte hashes, all summaries, wrong-row ledgers, mint ledgers, 4,800 raw sealed plans (mtimes preserved) and the bug audit are committed in results/v17. Repro commands in V1_7_FINAL_REPORT.md §11. Timing = harness-observed; no model-reported time or energy used anywhere.